

Optimality conditions: unconstrained problems

Necessary optimality conditions

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
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Necessary optimality conditions

We want to characterize an optimal solution

We remind classical results from analysis

Fermat's theorem

Theorem 5.1

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- ▶ x^* is a local minimum of $f : \mathbb{R}^n \rightarrow \mathbb{R}$.
- ▶ If f is differentiable around x^* , then

$$\nabla f(x^*) = 0.$$

- ▶ If f is twice differentiable around x^* , then

$$\nabla^2 f(x^*) \geq 0 \quad [\text{positive semidefinite}].$$

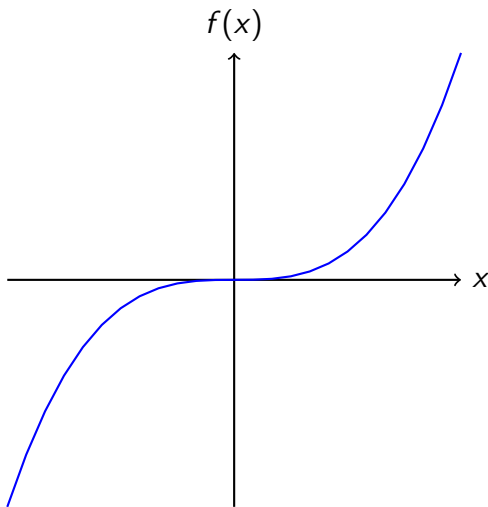
Necessary, but not sufficient

Example

$$f(x) = x^3$$

$$f'(x) = 3x^2, f''(x) = 6x$$

$$f'(0) = 0, f''(0) \geq 0$$



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- ▶ Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $\nabla^2 f(x^*)$
- ▶ $\nabla^2 f(x^*) \geq 0 \iff \lambda_i \geq 0, i = 1, \dots, n$
- ▶ Eigenvalue = curvature along the eigenvector.

Summary

At a local optimum, the gradient is null and the second derivative matrix is positive semidefinite. But these conditions are not sufficient.